Resolution refinement followed by the rodeo algorithm for eigenstate preparation: lattice refinement of the Hubbard model

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Background: Resolution refinement, the adiabatic transport of an eigenstate from a coarse to a fine model space along a shift-controlled, gapped path, was proposed for lattice problems in which every coarse site is resolved into a block of fine sites. Its implementation on the pairing Hamiltonian was validated at the gate level against alternative inputs to the same rodeo filter. **Purpose:** To carry the pipeline, the benchmarks and the cost accounting to the lattice refinement of the Hubbard model, and to establish what controls the refinement there and what it costs against the simplest alternatives. **Methods:** Every stage is benchmarked against exact diagonalization in the conserved $(N_\uparrow, N_\downarrow)$ sector for the open chain refined from four to eight and from eight to sixteen sites and for the $2 \times 2 \to 4 \times 4$ plaquette. The Galerkin coarse model is derived in closed form, the transported state is followed with total-spin and point-group diagnostics, the rodeo filter is evaluated with the exact Gaussian-ensemble formula, and all inputs — prolonged, refined along a projector path and along a two-local un-dimerization path, an ordinary adiabatic ramp from the free-fermion determinant, and full and pruned unitary coupled-cluster singles-and-doubles (UCCSD) states — are fed to the same second-order Trotterized filter and compared by ground-state infidelity against controlled-NOT (CNOT) count per accepted preparation. **Results:** The coarse model is again a Hubbard model, with $(t, U) \to (t/2, U/2)$ in one dimension, so the hierarchy is self-similar and can start from the filled dimer, a single basis state, reaching the eight- and sixteen-site ground states with overlap $p > 0.999$ without any variational input. The path conserves total spin and the lattice symmetry, so the transport is controlled by the lowest excitation in the symmetry sector of the target, $\Delta_{\text{eff}} = 0.37t$–$0.88t$, while the sector gap collapses to $0.14t$ (chain) and $0.05t$ (sixteen sites): the refinement time is flat in U and in size, and Δ_{eff} is the right width for the filter. A coarse state in the wrong representation has exactly zero overlap with the target (B_1 singlet versus A_2 triplet on the plaquette); refining the lowest coarse state of the right sector instead gives $p = 0.99997$. At the circuit level ($U = 4t$, sixteen qubits) the ordinary adiabatic ramp from the free-fermion determinant is the cheapest route at every target accuracy, 1.9×10^4, 4.0×10^4 and 9.1×10^4 CNOTs for infidelities 10^{-3}, 3×10^{-4} and 10^{-4}; the pruned 98-parameter UCCSD needs three times more, the two-local refinement path four to seven times more, and the projector path sixteen times more, almost entirely in the coarse term conditioned on all antibonding qubits. **Conclusions:** Lattice refinement of the Hubbard model is a validated, optimizer-free and measurement-free route to the fine ground state whose convergence is governed by a symmetry-protected gap, and the symmetry diagnostics identify in advance the coarse states that cannot be refined. At this size it is not a gate saver, however: an adiabatic ramp that switches on the interaction from the free-fermion determinant reaches the same accuracy at a fraction of the cost. The comparison fixes the baseline that larger lattices, where the free-fermion overlap decays, must beat.

I. INTRODUCTION

In Ref. [1] we implemented, end to end and at the gate level, a pipeline for eigenstate preparation in which an eigenstate of a coarse Hamiltonian is lifted by a prolongation operator into a finer model space, transported adiabatically along a shift-controlled, gapped path (*resolution refinement* [3]) and then cleaned by the rodeo algorithm [4, 5], a projective spectral filter. The testbed there was the constant-pairing Hamiltonian, exactly solvable by the Richardson ansatz, with "resolution" meaning the number of single-particle levels at fixed particle number, and the assessment was sober: at eight qubits the refinement stage was tied with simply filtering the Hartree–Fock determinant, cost 2.5–3 times an ordinary adiabatic

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ramp from the Hartree–Fock Hamiltonian at every target accuracy, and was beaten by a four-parameter pair-rotation ansatz; its merits were the absence of a classical optimization loop and of training measurements. A seniority-breaking particle-hole interaction, studied in the companion paper [2], left the structure and convergence of the pipeline unchanged while raising the depth of a Trotter step ninefold.

The present paper carries the same pipeline, the same benchmarks and the same cost accounting to the Hubbard model [6, 7],

$$H = -\sum_{\langle ij\rangle\sigma} t_{ij}\big(c^\dagger_{i\sigma}c_{j\sigma} + \text{h.c.}\big) + U\sum_i n_{i\uparrow}n_{i\downarrow}, \quad (1)$$

for three reasons. First, the Hubbard model is the canonical strongly correlated fermion problem and the most frequently used benchmark for quantum simulation of fermions [8–11]; it contains the one-body particle-hole channel (hopping) that the pairing model lacks and the two-body on-site term in their simplest forms, so it tests the lesson of Ref. [2] on a model where the seniority symmetry is absent from the outset. Second, it is the setting for which resolution refinement was proposed [3]: a *lattice* refinement in which every coarse site is resolved into a block of fine sites, the prolongation is a product of Givens rotations rather than the gate-free embedding of the basis refinement, and the fine lattice is the natural discretization of a continuum problem. Third, the model is not exactly solvable in general, so the exact-diagonalization (ED) reference that anchored every number in Ref. [1] must now be produced numerically; we keep that discipline and benchmark every stage against ED in the conserved $(N_\uparrow, N_\downarrow)$ sector with the dense and Lanczos solvers of our ED code [12], which also covers the t–J model and the chain, square and checkerboard lattices.

Two structural results make the lattice refinement of the Hubbard model particularly clean. The Galerkin projection of the fine Hubbard Hamiltonian onto the prolonged subspace is itself a Hubbard model, with $(t, U) \to (t/2, U/2)$ in one dimension — so the hierarchy of resolutions is self-similar at fixed U/t — and $(t, U) \to (t/2, U/4)$ in two dimensions (Sec. III B). And the interpolation path conserves total spin and the point-group symmetry of the lattice, so the gap that controls the adiabatic transport is the distance to the next state *in the symmetry sector of the target*, which stays of order t while the sector gap of the fine Hamiltonian collapses with U and with system size (Sec. IV B); the converse is that a coarse ground state in the wrong representation cannot be refined at all (Sec. IV E).

The paper follows the structure of Ref. [1]. Section II fixes the model, the sector encoding, the ED reference and the symmetry diagnostics. Section III defines the lattice prolongation, the Galerkin coarse Hamiltonian, the two refinement paths, the rodeo filter and the variational comparator. Section IV reports the sector-exact results for the open chain ($2 \to 4 \to 8 \to 16$ sites) and the square plaquette ($2\times2 \to 4\times4$). Section V gives the Pauli structure of the Hubbard chain in the Jordan–Wigner encoding, the cost of the prolongation and of the embedded coarse term, the Trotterized refinement, the ordinary adiabatic ramp and the pruned variational ansatz that serve as baselines, and the matched-observable gate budgets with their minimum cost at fixed accuracy. Section VI discusses the results and the outlook. All numbers are produced by a reproducible notebook built on a tested library (Sec. VI 0 h).

II. THE HUBBARD MODEL, ITS ED REFERENCE AND THE SYMMETRY DIAGNOSTICS

We use $t_{ij} = t = 1$ on nearest-neighbor bonds, open boundary conditions throughout, the chain of L sites and the $L \times L$ square lattice with site index $i = xL + y$. The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately, and all calculations are done in one such sector of dimension $\binom{N}{N_\uparrow}\binom{N}{N_\downarrow}$.

a. Encoding and ED. Spin-orbitals are Jordan–Wigner ordered in two blocks, the N up orbitals followed by the N down orbitals. With this ordering a down hop picks up the parity of all up electrons twice, which cancels, so the sector Hamiltonian factorizes exactly as

$$H = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow + U\,\mathrm{diag}(\text{double occupancy}), \quad (2)$$

where K_σ is the one-species hopping matrix on the sorted occupation masks of that species. Reference energies and eigenvectors are obtained with dense diagonalization up to a few thousand states and with ARPACK Lanczos beyond. The builder is validated against free fermions at $U = 0$, the analytic dimer $E_0 = \frac{1}{2}(U - \sqrt{U^2 + 16t^2})$, and the term decomposition used for product formulas in Sec. V; the sector dimensions encountered below range from 1 (the filled dimer) to 14 400 (four particles on sixteen sites).

b. Symmetry diagnostics. Three operators are built in the same sector basis and used to label every eigenstate. The total spin follows from $\langle \mathbf{S}^2 \rangle = S_z(S_z + 1) + \|S^+\psi\|^2$, with $S^+ = \sum_i c^\dagger_{i\uparrow}c_{i\downarrow}$ applied as a map from the $(N_\uparrow, N_\downarrow)$ to the $(N_\uparrow + 1, N_\downarrow - 1)$ sector with the Jordan–Wigner signs of the block ordering. A lattice symmetry is a fermionic site permutation $c^\dagger_i \to c^\dagger_{\pi(i)}$, i.e. a signed permutation matrix on each spin species; for the open chain this gives the reflection parity, and for the open square the eight elements of C_{4v}, from which the weight of a state in each irreducible representation follows from the character formula $w_\Gamma = (d_\Gamma/8)\sum_g \chi_\Gamma(g)^* \langle\psi|g|\psi\rangle$.

c. Why the Hubbard model after the pairing model. In the pairing model the seniority symmetry confines the ground state to a subspace in which the UCCSD solver is a doubles theory and the refinement path has large gaps; the particle-hole term of Ref. [2] broke that symmetry by hand. In Eq. (1) the hopping term is a one-body particle-hole operator from the outset, single excitations do not

vanish, and the only symmetries left to the pipeline are the ones it inherits from the lattice: particle number per spin, total spin S, and the point group. These are exactly the symmetries that, as we show, decide which gap governs the refinement.

III. THE PIPELINE FOR LATTICE REFINEMENT

A. Prolongation: coarse sites to blocks of fine sites

A refinement step maps a coarse lattice of N_c sites onto a fine lattice of $N_f = bN_c$ sites by resolving every coarse site c into a block $B(c)$ of b fine sites, $b = 2$ for the chain (sites $2c, 2c+1$) and $b = 4$ for the square lattice (2×2 plaquettes). The prolongation sends the coarse orbital to the normalized symmetric (bonding) combination of its block,

$$c^\dagger_{c\sigma} \;\longmapsto\; s^\dagger_{c\sigma} = \frac{1}{\sqrt{b}} \sum_{f \in B(c)} c^\dagger_{f\sigma}, \tag{3}$$

and a coarse Slater determinant to the corresponding product of $s^\dagger$ operators; particle numbers are unchanged, so the filling halves (quarters) at each step, as it must for a sequence of lattices approaching the continuum at fixed particle number [3]. The map P is an isometry, $P^\dagger P = \mathbb{1}$, from the coarse sector of dimension $\binom{N_c}{N_\uparrow}\binom{N_c}{N_\downarrow}$ into the fine sector; its complement is spanned by configurations in which at least one antibonding block mode (and, for $b = 4$, one of the two further orthogonal block modes) is occupied. On a quantum register in which the fine qubits of a block are adjacent, P is implemented by preparing the coarse state on one qubit per block per spin and applying one Givens rotation per block per spin — two CNOTs each, i.e. 16 CNOTs for $4 \to 8$ — in contrast to the gate-free embedding of the basis refinement. In the sector basis P is built explicitly: for each coarse occupation pattern the fine amplitudes are $b^{-n_\sigma/2}$ on every choice of one fine site per occupied block, with the sign of the permutation that sorts the chosen fine indices. Both $P^\dagger P = \mathbb{1}$ and the identities below are verified to machine precision in the notebook.

B. The Galerkin coarse Hamiltonian

The coarse Hamiltonian whose eigenstate is to be refined is the fine Hamiltonian seen at coarse resolution, $H_{\rm low} = P^\dagger H_{\rm high} P$. With this choice the identity $\langle P\psi_{\rm low}|H_{\rm high}|P\psi_{\rm low}\rangle = E_{\rm low}$ holds by construction for the exact coarse eigenstate, as in Ref. [1]; it says nothing about the overlap p_0 of the prolonged state with the fine ground state, which must be computed, and it is lost when the coarse state is itself an approximation with infidelity $\epsilon_{\rm low}$ (a variational coarse solver, or the output of a previous refinement step). In that case the coarse error propagates through the pipeline as in Sec. III of Ref. [1]: the final infidelity is bounded by $(\sqrt{\epsilon_{\rm low}} + a_{\rm ad} + \eta)^2$, with $a_{\rm ad}$ the adiabatic leakage of the transport and η the product-formula error, and Secs. IV C and IV G show that in practice the coarse admixture is transported almost unchanged and caps the refined overlap at roughly the coarse fidelity. The Galerkin model has a closed form. Projecting the fine hopping term with Eq. (3): an intra-block bond contributes $-t(n_s - n_a)$, i.e. a constant $-t$ per particle on the prolonged subspace, and each inter-block bond contributes $\frac{1}{2}s^\dagger_c s_{c'}$; projecting the on-site term, $n_{f\sigma} \to n_{s_c\sigma}/b$ for every $f \in B(c)$, gives $U\sum_{f\in B(c)} n_{f\uparrow}n_{f\downarrow} \to (U/b)\, n_{s_c\uparrow}n_{s_c\downarrow}$. Hence

$$\text{chain:}\quad H_{\rm low} = H_{\rm Hubbard}\big(L;\, t/2,\, U/2\big) - tN, \tag{4}$$

$$\text{square:}\quad H_{\rm low} = H_{\rm Hubbard}\big(L\times L;\, t/2,\, U/4\big) - 2tN, \tag{5}$$

with $N = N_\uparrow + N_\downarrow$, and with the same boundary conditions as the fine lattice. In one dimension the ratio U/t is preserved, so a chain of refinements $L \to 2L \to 4L \to \cdots$ is a sequence of Hubbard chains with the same interaction strength and a hopping scale that doubles at each step: the hierarchy is self-similar, and the coarsest member can be the two-site dimer or, for $N = 2L$, the filled dimer, a single computational basis state. In two dimensions the effective interaction of the coarse model is weaker by a factor of two relative to its hopping. We note that the Galerkin pair $(H_{\rm low}, H_{\rm high})$ is a choice; a continuum-scaling pair ($t \propto a^{-2}$, $U \propto a^{-1}$ in one dimension) would start the path from a different coarse ground state and change only the initial overlap p_0, not the construction.

C. Two refinement paths

a. Path A: the shift-controlled projector path. As in Ref. [1],

$$H_A(s) = \cos^2\!\big(\tfrac{\pi s}{2}\big) P(H_{\rm low} - \mu)P^\dagger + \sin^2\!\big(\tfrac{\pi s}{2}\big)(H_{\rm high} - \mu), \qquad s = t/T, \tag{6}$$

with the shift $\mu = E_{\rm low} + \mu_{\rm buf}$ placing the complement of the prolonged subspace (where $P(H_{\rm low} - \mu)P^\dagger$ vanishes) at energy $\mu_{\rm buf}$ above the tracked state at $s = 0$. Because $P(H_{\rm low} - \mu)P^\dagger = \Pi_a(H_{\rm high} - \mu)\Pi_a$ with Π_a the projector onto empty antibonding modes, the coarse term is, on a device, the fine Hamiltonian in the rotated (bonding/antibonding) frame conditioned on all antibonding qubits being in $|0\rangle$ — the multi-controlled structure whose cost dominated the $3 \to 4$ chain of Ref. [1], now with $2N_c$ control qubits.

b. Path B: adiabatic un-dimerization. The same end points are connected by a path that stays two-local in the laboratory frame: add to the fine Hamiltonian an extra hopping Λ on the intra-block bonds and switch it off,

$$H_B(s) = H_{\rm high} - \Lambda\cos^2\!\big(\tfrac{\pi s}{2}\big) \sum_{\langle ff'\rangle \in \text{intra}} \sum_\sigma \big(c^\dagger_{f\sigma}c_{f'\sigma} + \text{h.c.}\big). \tag{7}$$

At $s = 0$ the antibonding modes are pushed up by 2Λ, and for $\Lambda \to \infty$ the ground state of $H_B(0)$ is the prolonged Galerkin ground state; at finite Λ it differs from it by an admixture of order t/Λ, so a transport that starts from the prolonged state carries a residual $1 - p \simeq c\,(t/\Lambda)^2$ to the end. Path B is the dimerized Hubbard chain (the plaquette Hubbard model in two dimensions) being adiabatically made uniform; every Trotter step along it costs what a step of the fine Hamiltonian costs, with no conditioning, and the extra term only rescales the angles of the intra-block blocks.

c. Refinement time. The exact evolution along either path is computed as in Ref. [1]: the s interval is split into $M_s = 96$ slices, the Hamiltonian at the slice midpoint is diagonalized once, and the slice propagators are reused for every T. We report the overlap $p(T) = |\langle \Psi_{\rm high} | \psi(T) \rangle|^2$ with the exact fine ground state on a grid of T and quote T^* as the first grid value with $1 - p(T) < 10^{-3}$. For the 14 400-dimensional sectors the slice propagators are applied with a sparse Krylov exponential (48 slices). As in Ref. [1] the pipeline is optimizer-free in execution but not parameter-free: the refinement time T, the shift $\mu_{\rm buf}$ (or the penalty Λ), the number of product-formula steps n_s, the filter step δt and the objective energy $E_{\rm obj}$ are all chosen here with knowledge of the exact answer, and a device-level calibration protocol for them is part of the open work (Sec. VI).

D. The rodeo filter

The filter is unchanged from Ref. [1]: M cycles of the controlled evolution $e^{-iH_{\rm high}t_k}$ on one reused ancilla with the phase $e^{iE_{\rm obj}t_k}$, Gaussian cycle times of RMS width σ, and post-selection on the ancilla. For an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ and $E_{\rm obj} = E_0$ the fidelity of the ensemble of accepted runs obeys exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k\neq 0} |c_k|^2 q_k^M}{p + \sum_{k\neq 0} |c_k|^2 q_k^M}, \qquad q_k = \tfrac{1}{2}\big(1 + e^{-(E_k - E_0)^2\sigma^2/2}\big), \tag{8}$$

which we evaluate from the ED spectrum and quote as the reference value throughout. Equation (8) is the average over the Gaussian ensemble of cycle times (drawn afresh for every attempt) with exact controlled evolution and $E_{\rm obj}$ on resonance; the factor $q_k^M \geq 2^{-M}$ makes 2^{-M} a lower bound on the suppression of each excited component, not a rate that a single schedule attains, and a sampled average over a finite number of time sets, as in the circuit-level study of Sec. V C, scatters around Eq. (8) with a relative standard error that grows with M. The width is set by the smallest excitation energy that carries weight in the input, $\sigma = 5/\Delta_{\rm eff}$ (Sec. IV B); choosing it from the bare sector gap would lengthen the controlled evolutions by a factor of up to three for no gain.

E. The variational comparator

As in Ref. [1] the pipeline is compared with a unitary coupled-cluster singles-and-doubles (UCCSD) preparation. The reference is the free-fermion determinant (the N_σ lowest orbitals of the hopping matrix of the lattice), the pool contains the spin-resolved singles and doubles from it, and the state is the factorized (disentangled) product of exponentials $\prod_\mu e^{\theta_\mu(\tau_\mu - \tau_\mu^\dagger)} |{\rm HF}\rangle$ in a fixed order — all singles first, then the doubles, each class in lexicographic order of its orbital labels — optimized by L-BFGS with the analytic (adjoint) gradient; a different ordering is a different ansatz. The simulation is exact for the ansatz: the sector is transformed to the orbital occupation basis, where every generator connects pairs of determinants and its exponential is the Givens-type rotation $1 + \sin\theta\, K + (\cos\theta - 1)\Pi$, $\Pi = -K^2$, the same structure as the compiled excitation blocks of Ref. [1]. On a device the reference and the excitations live on the site register after a Givens network ($N(N-1)/2$ rotations per spin), and we count the circuit with the Pauli-rotation primitive of Sec. V: a single compiles into two strings, a double into eight, and a weight-w string costs $2(w-1)$ CNOTs. The comparator is used in two ways, as the coarse solver on the four-site chain (26 parameters, 8 qubits), whose state is then prolonged and refined, and as the direct competitor on the eight-site chain (198 parameters, 16 qubits). Since most of the 198 amplitudes are small, we also report the *pruned* ansatz obtained by keeping the operators whose converged amplitude exceeds 10^{-3} and re-optimizing: it is the demonstrated restricted ansatz that an adaptive construction such as ADAPT-VQE would at best find, and it is what the gate budget of Sec. V C should be compared with. Two further baselines are defined there: the free-fermion determinant itself, and the ordinary adiabatic ramp that switches on U from it.

IV. RESULTS

A. The open chain refined from four to eight sites

Table I and Fig. 1 summarize the refinement of the four-site chain with $N_\uparrow = N_\downarrow = 2$ (half filling, 36 states) to the eight-site chain (quarter filling, 784 states) for $U/t = 0, 2, 4, 8$. Three features stand out. (i) The prolonged state is a good start at every coupling, $p_0 = 0.75$–0.81, with a weak maximum at $U = 2t$; unlike the pairing model, the free-fermion case is not trivial, because the bonding modes are not the lowest fine single-particle orbitals. (ii) The sector gap of the fine Hamiltonian falls by a factor of four over the range of U, from $0.53t$ to $0.14t$, but the refinement time does not move: with $\mu_{\rm buf} = 2t$ the overlap reaches $p > 0.997$ at $T = 8/t$, $1 - p < 10^{-3}$ at $T^* = 10/t$ and $p > 0.9999$ at $T = 16/t$ for all four couplings, and with $\mu_{\rm buf} = t$ the same values are reached

at roughly twice the time. The minimum of the sector gap along Path A sits at $s = 1$ (it is the fine-sector gap itself) in every case, so the global path gap cannot be the quantity that controls the transport; the reason is the symmetry protection discussed next. (iii) Path B refines at a comparable rate but saturates at the finite-Λ ceiling: at $T = 64/t$, $1 - p = 4.8$–7.3×10^{-3} for $\Lambda = 8t$ and 1.7–3.7×10^{-3} for $\Lambda = 16t$, with a weak and non-monotonic U dependence, consistent with the $(t/\Lambda)^2$ scaling of the admixture (the overlap of the prolonged state with the exact ground state of $H_B(0)$ is 0.9944–0.9953 at $\Lambda = 8t$ and 0.9983–0.9986 at $\Lambda = 16t$). The ceiling, not the transport, limits Path B; a schedule with $\Lambda \to \infty$ at $s = 0$ or a final filter removes it.

B. Which gap controls the refinement: symmetry protection

Both terms of Eq. (6) commute with the total spin and with the point group of the fine lattice (the block structure respects the reflection of the chain and the C_{4v} group of the square), and the prolongation maps irreducible representations onto themselves. The transported state therefore never leaves the symmetry sector of the coarse ground state — a spin singlet, even under reflection, for the cases of Table I — and the states that can be excited along the path are only those of the same sector. Figure 2 makes this explicit with the symmetry labels of Sec. II: for $U = 4t$ the lowest fine excitations at $0.223t$ (a reflection-odd triplet) and $0.459t$ (an even triplet) carry weights of order 10^{-31}, and the first state with nonzero weight (4.6×10^{-5}), an even singlet, lies at $\Delta_{\rm eff} = 0.588t$; at $U = 8t$ the corresponding numbers are $0.137t$ and $0.286t$ (triplets, zero weight) and $\Delta_{\rm eff} = 0.366t$ (weight 3×10^{-7}), with the bulk of the excited weight lying still higher. The weight of the prolonged state in the even-singlet sector is one to twelve digits for every $U > 0$. $\Delta_{\rm eff}$ falls with U much more slowly than Δ, which explains the U-independent T^* of Table I.

Figure 4 follows the two gaps along Path A: the sector gap and the gap to the lowest state with the same (S, parity) as the tracked state. The latter never approaches the former; its minimum is $\Delta_{\rm eff}$ itself, at $s = 1$. This is the isolated-branch condition of the adiabatic theorem verified numerically along the whole path, within the symmetry sector in which the transport actually takes place, rather than assumed. Table II evaluates the sufficient adiabatic time of Jansen, Ruskai and Seiler [13] in the form used in Sec. IV.E of Ref. [1], $T_b = \epsilon^{-1/2} \int_0^1 [\|\ddot{H}\|/\Delta^2 + 7\|\dot{H}\|^2/\Delta^3]\, ds$ with $\|B - A\|$ the spectral norm of $H_{\rm high} - P H_{\rm low} P^\dagger$ in the sector, once with the sector gap and once with the symmetry-resolved gap. The symmetry-resolved bound is smaller by a factor 5 to 17 but remains three to five orders of magnitude above the empirical $T^* = 10/t$, as the pairing bound did (there 3×10^2); the larger ratio here reflects $\|B - A\| = 11$–20, set by the shift μ and the width of the Hubbard spectrum. As in Ref. [1] the two numbers answer different questions — the bound guarantees a worst-case operator-norm error for every initial state in the branch, T^* records one overlap for one initial state — so the ratio measures the slack of the criterion, not an error in either.

The first-order response norm of the path, $\chi_u(s) = \sum_{n>0} |\langle n(s)|\partial_u H|0(s)\rangle|^2/[E_n(s) - E_0(s)]^2$ with $u = \sin^2(\pi s/2)$ and $\partial_u H = (H_{\rm high} - \mu) - P(H_{\rm low} - \mu)P^\dagger$, is the quantity whose integral along the path sets the adiabatic leakage at leading order; for the pairing model Ref. [1] obtained it in closed form at the start of the path, $\chi_u(0) = g^2/\mu_{\rm buf}^2$. Figure 3 (left) follows it along Path A for the chain $4 \to 8$: $\chi_u(0) = 0.375$, 0.527, 0.715 and 0.874 for $U/t = 0, 2, 4, 8$, the maximum sits at the initial endpoint for every coupling, and the norm falls by a factor two to four along the path (at $U = 8t$ it rises again toward $s = 1$, to 0.45, as $\Delta_{\rm eff}$ shrinks). The shift fixes the initial value as it did for the pairing model, and the growth of $\chi_u(0)$ with U is modest, in line with the flat T^*.

C. Bootstrapping from the filled dimer and composing steps

The self-similarity of the one-dimensional hierarchy allows the chain of refinements to start from the two-site dimer with $N_\uparrow = N_\downarrow = 2$, whose sector contains a single state — both sites doubly occupied — with coarse energy $E_{\rm low} = 2 \cdot (U/2) - 2t \cdot 2 = U - 4t$. This is the analog of the $k = 2$ determinant that seeded the $k = 2 \to 4$ chain in Ref. [1], and it requires no classical or variational preprocessing. Table III shows the step $2 \to 4$: the doubly-occupied seed is, as expected, a worse start as U grows, $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but the effective gap is large ($\Delta_{\rm eff} = 2.2t$–$0.8t$) and $p > 0.9997$ is reached by $T = 16/t$ at every U.

Figure 5 composes the steps. With the *refined* four-site state (not the exact one) as the input of the $4 \to 8$ step and the shift referenced to its variational energy $\langle\psi|H_{\rm low}|\psi\rangle$, as in the $3 \to 4$ chain of Ref. [1], the composed chain $2 \to 4 \to 8$ at $U = 4t$ with $T = 16/t$ per step reaches $p = 0.99988$, against 0.99994 from the exact coarse input; with $T = 8/t$ in the first step the second step saturates at $p = 0.9979$, the overlap of its input, because the admixture of other coarse states refines to other fine states. The composed chain is also better than a single-shot refinement $2 \to 8$ in which each coarse site is resolved into a block of four at once: that start has $p_0 = 0.579$ and reaches $p = 0.9966$ at $T = 16/t$ and 0.99988 only at $T = 32/t$. Two steps of $T = 16/t$ therefore take a single computational basis state to the eight-site ground state with $p > 0.9998$ before any filtering.

TABLE I. Refinement of the open Hubbard chain from $L = 4$ to $L = 8$ sites at $N_\uparrow = N_\downarrow = 2$ ($t = 1$; 36 and 784 states). $E_{\rm low}$ is the ground-state energy of the Galerkin coarse model, Eq. (4), $E_{\rm high}$ the ED ground-state energy of the fine chain, p_0 the weight of the prolonged coarse ground state on the fine ground state, Δ the sector gap of the fine Hamiltonian and $\Delta_{\rm eff}$ the lowest excitation energy carrying weight in the prolonged state. $p(T)$ is the overlap after refinement along Path A ($\mu_{\rm buf} = 2t$) and Path B ($\Lambda = 16t$); the last columns give the ensemble infidelity, Eq. (8), of the rodeo filter with $\sigma = 5/\Delta_{\rm eff}$ applied to the prolonged state and to the Path-A state refined for $T = 16/t$.

U/t	$E_{\rm low}$	$E_{\rm high}$	p_0	Δ	$\Delta_{\rm eff}$	$p_A(8)$	$p_A(16)$	$p_B(16)$	$p_B(32)$	$1-\mathcal{F}_M$, prolonged		$1-\mathcal{F}_M$, refined T=16	
										$M=2$	$M=6$	$M=2$	$M=6$
0	−6.236068	−6.822948	0.777	0.532	0.879	0.99718	0.99997	0.99660	0.99838	6.7×10^{-2}	4.5×10^{-3}	8.6×10^{-6}	5.4×10^{-7}
2	−5.437971	−6.032090	0.814	0.319	0.780	0.99876	0.99996	0.99594	0.99849	5.4×10^{-2}	3.6×10^{-3}	9.4×10^{-6}	5.9×10^{-7}
4	−4.976573	−5.634427	0.797	0.223	0.588	0.99837	0.99994	0.99316	0.99769	6.0×10^{-2}	4.0×10^{-3}	1.5×10^{-5}	9.2×10^{-7}
8	−4.558586	−5.277973	0.750	0.137	0.366	0.99761	0.99991	0.99205	0.99731	7.7×10^{-2}	5.2×10^{-3}	2.2×10^{-5}	1.4×10^{-6}

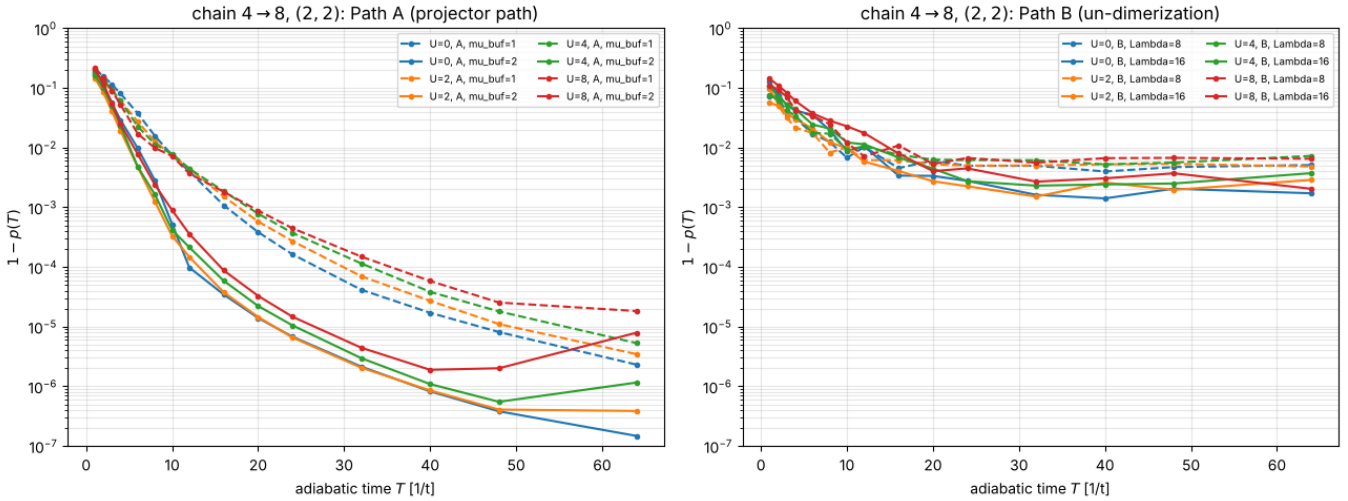


FIG. 1. Refinement of the open chain $4 \to 8$ at $(2,2)$: $1-p(T)$ along Path A (left; solid $\mu_{\rm buf} = 2t$, dashed $\mu_{\rm buf} = t$) and along Path B (right; solid $\Lambda = 16t$, dashed $\Lambda = 8t$) for $U/t = 0, 2, 4, 8$. Path A reaches $1-p < 10^{-4}$ by $T = 16/t$ at every coupling; Path B saturates at the finite-Λ ceiling of a few 10^{-3}.

TABLE II. Sufficient adiabatic time T_b (infidelity $\epsilon = 10^{-2}$) along Path A ($\mu_{\rm buf} = 2t$) for the chain $4 \to 8$, evaluated with the sector gap and with the gap in the symmetry sector of the tracked state, against the empirical T^* ($1-p < 10^{-3}$). Times in units of $1/t$.

U/t	$\min\Delta$	$\min\Delta_{\rm eff}$	$\|B-A\|$	$T_b(\Delta)$	$T_b(\Delta_{\rm eff})$	T^*
0	0.532	0.879	11.1	5.7×10^4	1.2×10^4	10
2	0.319	0.780	11.8	2.5×10^5	2.0×10^4	10
4	0.223	0.588	13.9	1.0×10^6	6.4×10^4	10
8	0.137	0.366	20.0	1.0×10^7	5.8×10^5	10

D. Eight to sixteen sites

The protection survives a doubling of the system. Refining the eight-site chain to sixteen sites at $(2,2)$ ($784 \to 14\,400$ states, $U = 4t$) starts from $p_0 = 0.869$, with $E_{\rm low} = -6.817213$ and $E_{\rm high} = -7.209813$, faces a fine-sector gap of only $0.051t$, and still reaches $p = 0.9924$ at $T = 8/t$, 0.99931 at $T = 16/t$ and 0.99995 at $T = 32/t$. Among the forty lowest eigenstates of the fine sector the first to carry weight lies at $\Delta_{\rm eff} = 0.138t$ (weight 10^{-4}), the next at $0.415t$ and $0.505t$ (the latter with weight 10^{-2}), while the eight other states below $0.4t$, including the one that defines the sector gap, carry none. Composing three steps from the filled dimer, $2 \to 4 \to 8 \to 16$ with $T = 16/t$ each and the refined state as the input of

TABLE III. The first step of the one-dimensional hierarchy: the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state) refined to the four-site open chain (36 states), Path A with $\mu_{\rm buf} = 2t$.

U/t	$E_{\rm low}$	$E_{\rm high}$	p_0	Δ	$\Delta_{\rm eff}$	$p(8)$	$p(16)$
0	−4	−4.4721	0.805	1.236	2.236	0.9999	1.0000
2	−2	−2.8759	0.754	0.789	1.851	0.9995	1.0000
4	0	−1.9531	0.645	0.540	1.358	0.9979	0.9999
8	4	−1.1172	0.480	0.311	0.805	0.9787	0.9997

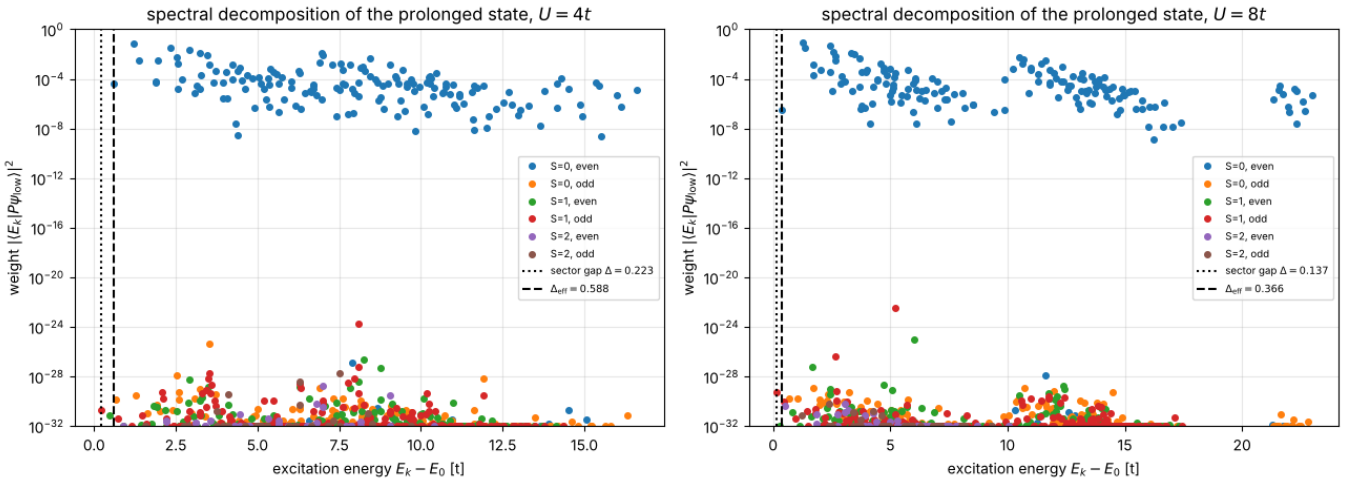


FIG. 2. Spectral decomposition of the prolonged coarse ground state over the eigenstates of the eight-site chain at $U = 4t$ (left) and $8t$ (right), each eigenstate labeled by its total spin and reflection parity. Only even singlets carry weight; the triplets and odd states, including the ones that define the sector gap Δ (dotted), sit at round-off level. The lowest state with weight defines $\Delta_{\rm eff}$ (dashed).

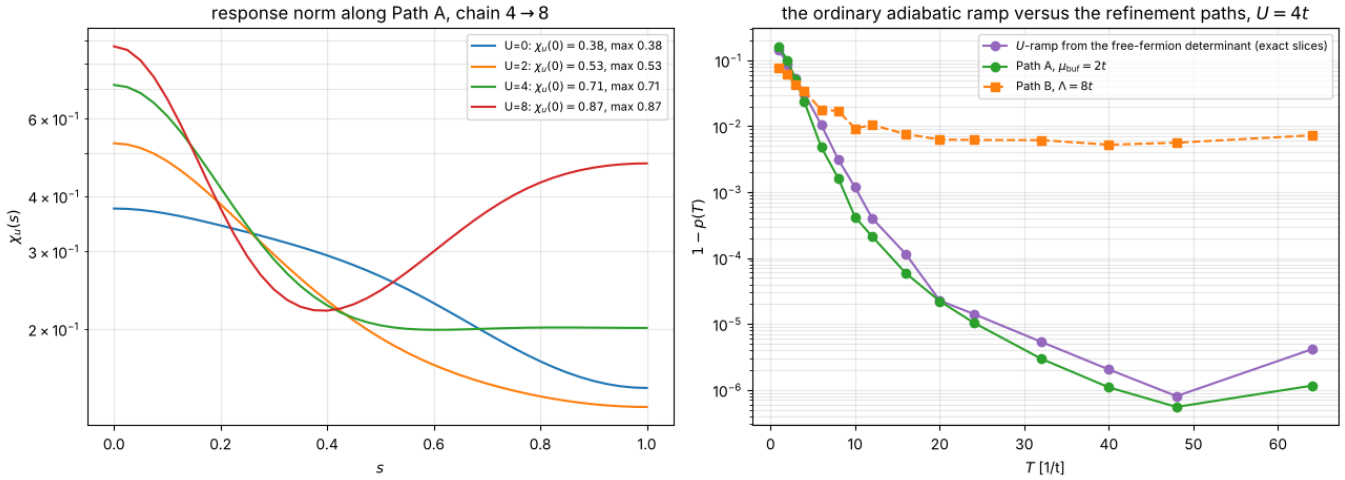


FIG. 3. Left: the first-order response norm $\chi_u(s)$ along Path A ($\mu_{\rm buf} = 2t$) for the chain $4 \to 8$ at $U/t = 0, 2, 4, 8$; the maximum sits at the initial endpoint for every coupling. Right: $1 - p(T)$ with exact slices for the ordinary adiabatic ramp from the free-fermion determinant, for Path A and for Path B ($\Lambda = 8t$) at $U = 4t$.

every step, gives $p = 0.99923$ for the sixteen-site ground state.

E. Two dimensions: the plaquette and the wrong symmetry sector

For the square lattice the first step is $2 \times 2 \to 4 \times 4$. With one particle per spin ($16 \to 256$ states) the refinement behaves as in one dimension: both ground states are A_1, $p_0 = 0.80$–0.85 for $U/t = 0, 4, 8$, $\Delta_{\rm eff} \simeq 2.1t$–$2.2t$, and $p > 0.9997$ at $T = 8/t$, $p > 0.99998$ at $T = 16/t$ (Table IV).

With two particles per spin ($36 \to 14\,400$ states) the prolonged coarse ground state has *zero* overlap with the fine ground state, and the symmetry diagnostics of Sec. II identify why (Fig. 6). The half-filled coarse plaquette (Galerkin parameters $t/2$, $U/4$) has, for every $U > 0$, a B_1 singlet ground state with an A_2 triplet close above it ($0.071t$ at $U = 4t$, $0.15t$ at $8t$); the dilute fine lattice — four electrons on sixteen open sites — has an A_2 *triplet* ground state for every $U > 0$ studied ($U/t = 1, 2, 4, 6, 8$), the B_1 singlet lying $0.04t$–$0.11t$ above it. The prolongation commutes with $\mathbf{S}^2$ and with C_{4v}, so the two cannot be connected: the decomposition of the fine ground state over the prolonged coarse eigenstates gives weight 0 on the coarse ground state, 0.647 on the A_2 triplet, and 0.007 on all other coarse states together ($U = 4t$). Refin-

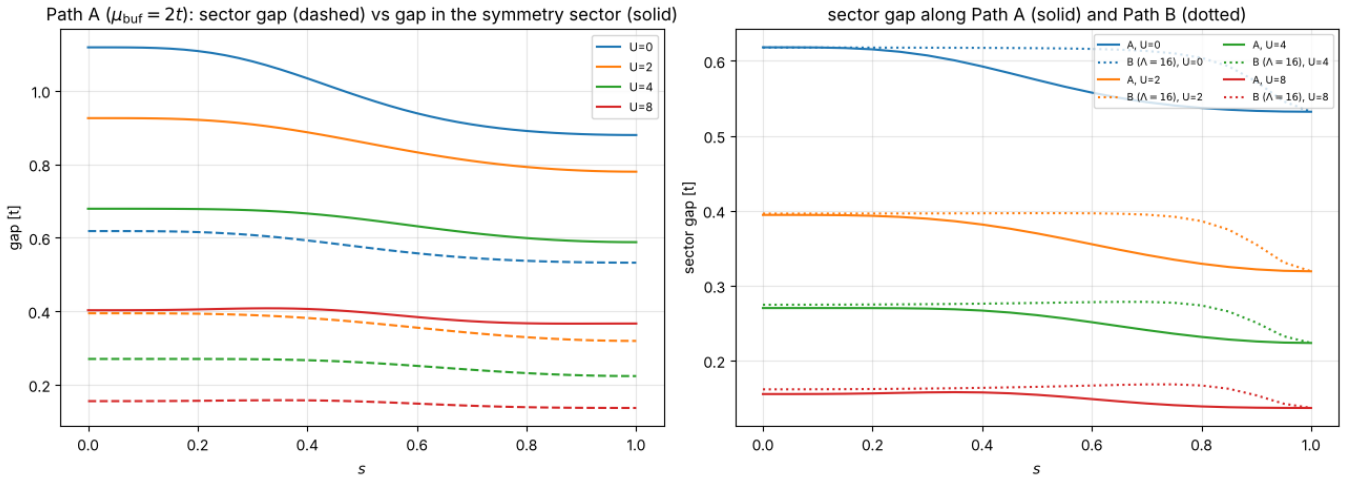


FIG. 4. Left: along Path A ($\mu_{\rm buf} = 2t$), the sector gap (dashed) and the gap within the symmetry sector of the tracked state (solid) for $U/t = 0, 2, 4, 8$. Right: the sector gap along Path A (solid) and Path B with $\Lambda = 16t$ (dotted); both reach their minimum, the fine-sector gap, at $s = 1$.

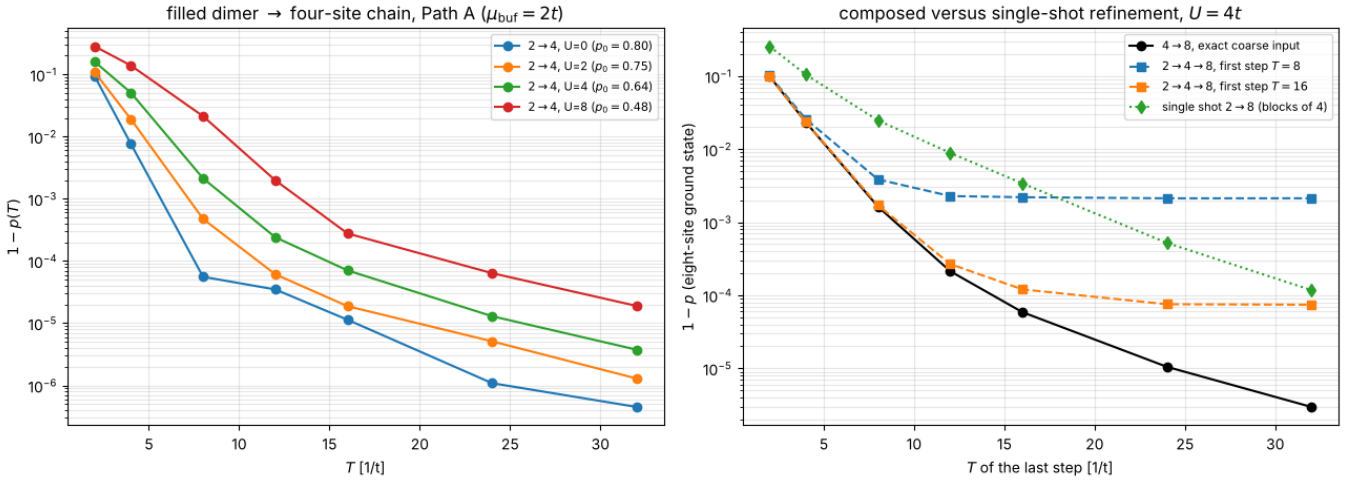


FIG. 5. Left: $1 - p(T)$ for the dimer step $2 \to 4$ at $U/t = 0, 2, 4, 8$. Right, $U = 4t$: the step $4 \to 8$ from the exact coarse ground state (circles), the composed chain $2 \to 4 \to 8$ with the refined four-site state as input after a first step of $T = 8/t$ or $16/t$ (squares), and the single-shot refinement $2 \to 8$ with blocks of four fine sites (diamonds), all versus the time of the last step.

ing that triplet along Path A ($\mu = E_1^{\rm low} + 2t$) reaches $p = 0.99944$ at $T = 8/t$, 0.99997 at $T = 16/t$ and 0.99999 at $T = 32/t$, although the fine-sector gap is only $0.090t$; the lowest state of the $(A_2, S = 1)$ sector that carries weight lies $1.77t$ above the ground state (among the thirty lowest eigenstates), so once the right sector is chosen the symmetry protection of Sec. IV B works strongly in its favor. The same path started from the coarse ground state gives $p = 3 \times 10^{-27}$.

Two conclusions follow for the method. The coarse state to be refined is the lowest coarse eigenstate *in the symmetry sector of the target*, which need not be the coarse ground state, and the coarse solver (UCCSD or otherwise) must therefore be run with the target's quantum numbers imposed or at least diagnosed; and the rodeo filter cannot rescue a wrong choice, since a state of zero overlap has nothing to amplify. Conversely, when the coarse ground state is in the wrong sector the refinement becomes a sensitive detector of a level crossing between the coarse and the fine problem — here between the singlet and the triplet as the filling goes from one half to one eighth, a crossing that persists over the whole range of U studied rather than being tuned by it. [The $U = 0$ point is degenerate in both models and its labels are not meaningful; consider adding the (S, Γ)-resolved spectrum of the fine lattice at a second filling, e.g. $(3, 3)$, and the periodic lattice, where Lieb's theorem fixes the spin at half filling.]

TABLE IV. Plaquette refinement $2 \times 2 \to 4 \times 4$ on the open square lattice, Path A with $\mu_{\rm buf} = 2t$. At $(2,2)$ the coarse ground state is a B_1 singlet with zero overlap on the A_2 triplet ground state of the fine lattice; the entry refines the lowest coarse $(A_2, S = 1)$ state (see text), and $\Delta_{\rm eff}$ is taken among the thirty lowest fine eigenstates.

$(N_\uparrow, N_\downarrow)$	U/t	dims	$E_{\rm low}$	$E_{\rm high}$	(Γ, S) coarse / fine	p_0	Δ	$\Delta_{\rm eff}$	$p(16)$
(1,1)	0	$16 \to 256$	-6.0000	-6.4721	A_1 / A_1	0.805	1.000	2.236	0.99999
(1,1)	4	$16 \to 256$	-5.8136	-6.2638	A_1 / A_1	0.840	0.792	2.096	0.99999
(1,1)	8	$16 \to 256$	-5.7093	-6.1833	A_1 / A_1	0.845	0.711	2.129	0.99998
(2,2)	4	$36 \to 14\,400$	-9.4142	-10.4197	$(B_1, 0)$ / $(A_2, 1)$	0	0.090	—	3×10^{-27}
(2,2)	4	$36 \to 14\,400$	-9.3429	-10.4197	$(A_2, 1)$ / $(A_2, 1)$	0.647	0.090	1.771	0.99997

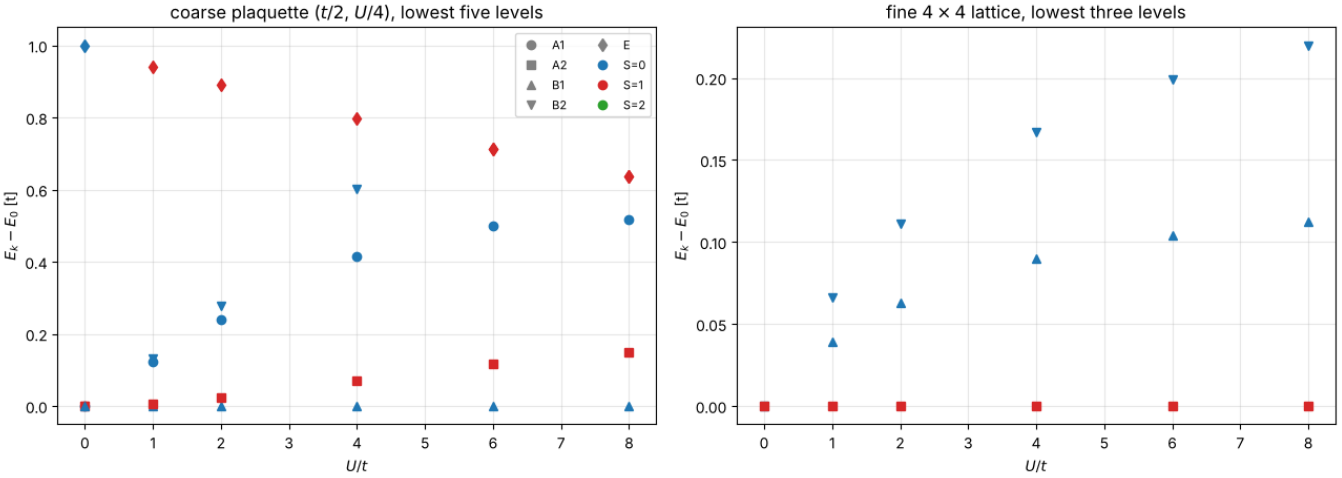


FIG. 6. Lowest levels of the coarse plaquette (left, Galerkin parameters $t/2$, $U/4$) and of the fine 4×4 lattice (right) at $(2,2)$ versus U/t, labeled by C_{4v} irrep (marker) and total spin (color). For every $U > 0$ the coarse ground state is a B_1 singlet and the fine ground state an A_2 triplet; the prolongation conserves both labels.

F. Rodeo filtering of the refined state

With $\sigma = 5/\Delta_{\rm eff}$ ($5.7/t$ to $13.6/t$ for $U/t = 0$ to 8) the filter applied to the prolonged state behaves as the pairing case did (Fig. 7): the infidelity falls from $1 - p_0 \simeq 0.2$ to 4–5×10^{-3} after six cycles and 2–3×10^{-4} after ten, the geometric suppression of Eq. (8) (at least 2^{-M} per excited component), with an ensemble acceptance that converges to p_0. Applied to the Path-A state refined for $T = 16/t$ the same filter starts from $1 - p \simeq 10^{-4}$ and reaches 0.5–1.4×10^{-6} after six cycles; applied to the Path-B state ($\Lambda = 16t$, $T = 32/t$) it starts from the ceiling, $1 - p \simeq 2 \times 10^{-3}$, and reaches 2–4×10^{-5} after six cycles; applied to the eight-site UCCSD state it starts from $1 - p = 4.7 \times 10^{-4}$, 3.1×10^{-3} and 1.2×10^{-2} at $U/t = 2, 4, 8$ (zero at $U = 0$, where the free-fermion reference is exact) and reaches 7×10^{-6}, 5×10^{-5} and 2×10^{-4}. As in Ref. [1], the question that matters is not whether the refined input converges faster per cycle — it does, by the ratio of the initial infidelities — but whether the cycles saved pay for the refinement gates; that comparison requires the circuit-level Trotterization of Sec. V.

G. The UCCSD comparator

Table V collects the variational results. On the coarse chain the 26-parameter UCCSD is exact at $U = 0$ and reaches infidelities 1.8×10^{-4}, 3.3×10^{-3} and 1.7×10^{-2} at $U/t = 2, 4, 8$ from a free-fermion reference that is very poor at strong coupling ($E_{\rm HF} - E_{\rm FCI} = 2.3t$ at $U = 8t$); on the fine chain the 198-parameter ansatz does slightly better, 4.7×10^{-4}, 3.1×10^{-3}, 1.2×10^{-2}, at a cost of 3.1×10^4 CNOTs against 1.7×10^3. Used as the coarse solver, the UCCSD state is prolonged with essentially the same p_0 as the exact coarse state (0.800 against 0.797 at $U = 4t$), but the refinement then saturates at $p = 0.99669$ ($U = 4t$) and 0.98234 ($U = 8t$) instead of 0.9999: as in the composed chain of Sec. IV C, the admixture of other coarse eigenstates in the variational input is transported to other fine eigenstates, and the refined overlap is capped at roughly the coarse fidelity. The coarse solver must therefore be converged to the accuracy required of the final state, or the filter must do the rest.

Most of the fine-chain amplitudes are small: at $U = 4t$, 72 of the 198 exceed 10^{-2} and 98 exceed 10^{-3}. Keeping those 98 operators (12 singles and 86 doubles)

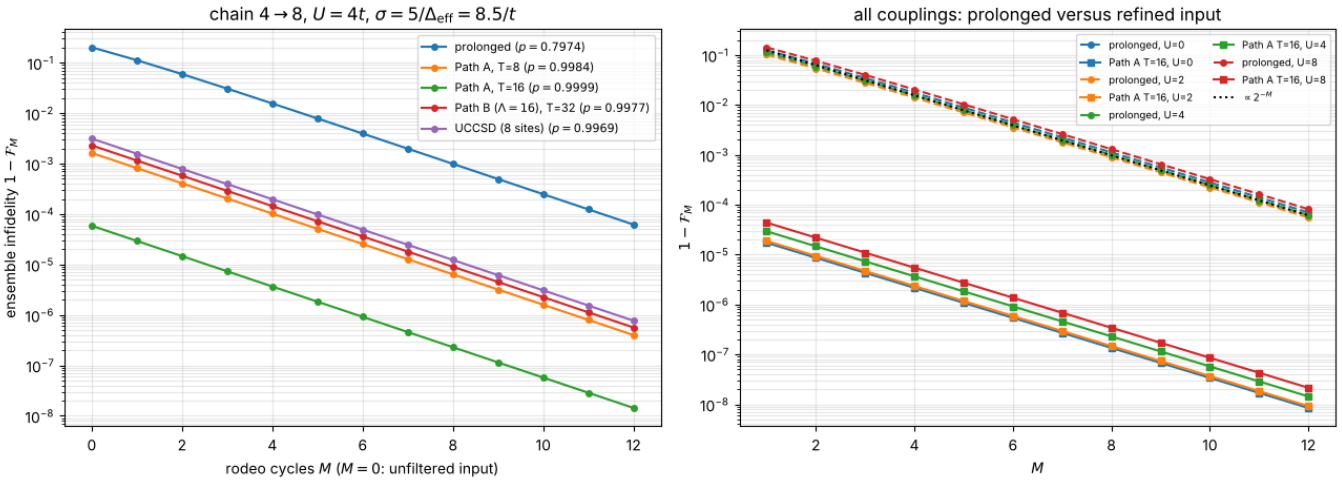


FIG. 7. Exact ensemble infidelity of the rodeo filter, Eq. (8), with $\sigma = 5/\Delta_{\rm eff}$ versus the number of cycles. Left, $U = 4t$: prolonged input, Path-A inputs refined for $T = 8/t$ and $16/t$, the Path-B input at its ceiling and the eight-site UCCSD state. Right: prolonged (dashed) and Path-A $T = 16/t$ (solid) inputs for all four couplings against the 2^{-M} law.

and re-optimizing from the converged values gives $E = -5.618755$ and $1 - F = 3.145 \times 10^{-3}$ against -5.618758 and 3.143×10^{-3} for the full pool: the same accuracy at half the gate count, 15 120 against 30 864 CNOTs. This pruned ansatz is the variational comparator of Sec. V C; it is a demonstrated restricted ansatz, not an adaptive selection, which would start from a different (and possibly smaller) operator set.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

A. Pauli structure and step costs

In the block Jordan–Wigner ordering of Sec. II an open chain of L_f sites on $2L_f$ qubits has $4(L_f - 1)$ hopping strings $\frac{t}{2}(X_jX_{j+1}+Y_jY_{j+1})$ of weight two, L_f interaction strings $\frac{U}{4}Z_jZ_{j+L_f}$ and $2L_f$ single-Z terms: 24 strings for $L_f = 4$, which we verified against the full-space Pauli decomposition of the library on eight qubits, and 52 strings of weight ≤ 2 for $L_f = 8$, against the 325 strings of weight up to eight of the $k = 4$ pairing model on half the qubits. With the Pauli-rotation primitive of Ref. [1] — $2(w - 1)$ CNOTs per weight-w string, two more per rotation when controlled — a first-order Trotter step of the fine chain costs 72 CNOTs (52 rotations), a second-order step 144, and a controlled second-order step 352. The product formula groups the bonds into two layers of disjoint bonds plus the diagonal interaction; each layer is a product of commuting two-qubit blocks and is realized exactly, so the Trotter error comes only from the non-commutativity between the three groups, as on hardware. The $Z_jZ_{j+L_f}$ terms are the only non-local ones; the site-major ordering, which makes them adjacent at the price of weight-three hopping strings, is the alternative to be compared. [Compare the two orderings on a linear-connectivity device.]

a. Prolongation and the coarse term. The prolongation $4 \to 8$ costs 16 CNOTs. The embedded coarse term of Path A is the coarse Hubbard step on the bonding register — 24 rotations, 32 CNOTs unconditioned — with every rotation conditioned on the eight antibonding qubits being empty. With the Gray-code phase-polynomial convention that gave the 14- and 350-CNOT figures of Ref. [1] (which is exact for few controls but grows as 2^{c+1}) this is 1.2×10^4 CNOTs per application; with a clean-ancilla ladder of relative-phase Toffolis ($6(c - 1) + 2$ extra CNOTs per rotation) it is 1 088. We use the latter, i.e. 7.6 fine Trotter steps per application of the coarse term, and note that either figure makes the coarse term, not the fine step, the cost of Path A. [Compile the conditioned coarse term explicitly and verify it against the sector matrix exponential, as was done for the $2 \to 4$ phase in paper I.]

B. Trotterized refinement

Figure 8 replaces the exact slice propagation at $T = 16/t$, $U = 4t$ by first- and second-order product formulas over the groups $[P(H_{\rm low} - \mu)P^\dagger, K_{\rm even}, K_{\rm odd}, D - \mu]$ for Path A and $[(1 + \Lambda\cos^2)K_{\rm even}, K_{\rm odd}, D]$ for Path B. The Hubbard path needs many more steps than the pairing path did: the second-order formula reproduces the exact-slice overlap to within 10^{-3} with $n_s = 128$ steps for Path A ($p = 0.99983$), 256 for Path B with $\Lambda = 8t$ (0.99213) and 512 for $\Lambda = 16t$ (0.99342), against 43 steps for $T^* = 16$ in Ref. [1]. The reason is the norm of the step generators, $\|B - A\| = 14$ for Path A and the enhanced intra-block hopping $(1 + \Lambda)$ for Path B, which makes $\Lambda = 8t$ the better operating point of the two-local

TABLE V. Sector-level UCCSD (Sec. III E) on the Galerkin coarse chain ($L = 4$, Eq. (4)) and on the fine chain ($L = 8$) at $(2, 2)$: reference, variational and exact energies, infidelity with the exact ground state, number of parameters and CNOT count of the preparation (Givens network plus excitation blocks). The last columns give the overlap of the prolonged UCCSD coarse state and its Path-A refinement at $T = 16/t$.

	coarse $L = 4$ (26 parameters, 1720 CNOTs)				fine $L = 8$ (198 parameters, 30864 CNOTs)				UCCSD coarse → refined	
U/t	E_{HF}	E_{UCCSD}	E_{FCI}	$1 - F$	E_{HF}	E_{UCCSD}	E_{FCI}	$1 - F$	p_0	$p(16)$
0	−6.23607	−6.23607	−6.23607	$< 10^{-15}$	−6.82295	−6.82295	−6.82295	$< 10^{-14}$	0.77733	0.99997
2	−5.23607	−5.43744	−5.43797	1.8×10^{-4}	−5.71184	−6.02962	−6.03209	4.7×10^{-4}	0.81405	0.99977
4	−4.23607	−4.96957	−4.97657	3.3×10^{-3}	−4.60073	−5.61876	−5.63443	3.1×10^{-3}	0.79991	0.99669
8	−2.23607	−4.51765	−4.55859	1.7×10^{-2}	−2.37850	−5.22167	−5.27797	1.2×10^{-2}	0.75667	0.98234

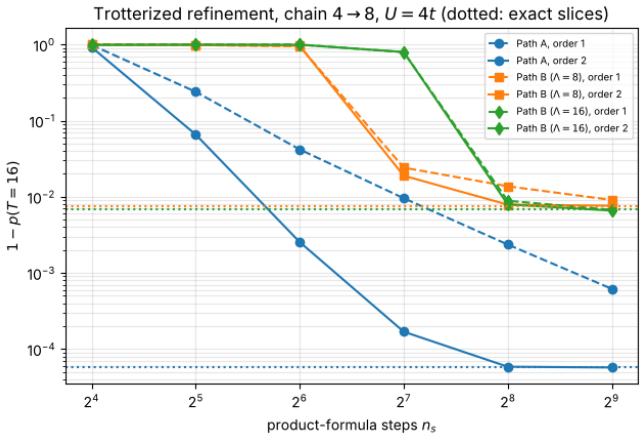


FIG. 8. Trotterized refinement of the chain $4 \to 8$ at $U = 4t$, $T = 16/t$: $1 - p$ versus the number of product-formula steps for Path A and Path B ($\Lambda = 8t$, $16t$), first (dashed) and second order (solid); dotted lines are the exact-slice values.

path despite its higher ceiling. At these step counts the refinement costs 2.97×10^5 CNOTs for Path A (of which 2.8×10^5 in the coarse term), 3.7×10^4 for Path B at $\Lambda = 8t$ and 7.4×10^4 at $\Lambda = 16t$ (Table VI). First-order formulas do not reach the exact-slice overlap within 10^{-3} even at 512 steps.

The counted circuit of Path A does not exponentiate the embedded coarse term as one block: on a device it is the coarse Hubbard step, i.e. the three conditioned coarse groups $P(D_c - \mu)P^\dagger$, $PK^c_{\mathrm{even}}P^\dagger$ and $PK^c_{\mathrm{odd}}P^\dagger$, which do not commute. Repeating the second-order refinement with the coarse term split accordingly (six groups in all) gives $p = 0.997299$, 0.999818, 0.999941 and 0.999943 at $n_s = 64$, 128, 256 and 512, against 0.997456, 0.999831, 0.999941 and 0.999943 with the coarse term as one block; the step count $n_s = 128$ and the cost of Path A are unchanged, and the split-group state is the one used below.

a. Baseline: the ordinary adiabatic ramp. The simplest adiabatic preparation of the fine ground state needs neither a coarse model nor a prolongation: start from the free-fermion determinant of the fine chain, prepared by a Givens network (2×28 rotations, 112 CNOTs), and switch on the interaction, $H_R(s) = K + \sin^2(\pi s/2)\, U \sum_i n_{i\uparrow} n_{i\downarrow}$. At $U = 4t$ the determinant has $p_0 = 0.7804$ — essentially the overlap of the prolonged state, 0.7974 — and the same $\Delta_{\mathrm{eff}} = 0.588t$, since it lies in the same symmetry sector. Figure 3 (right) compares the two transports with exact slices: the ramp reaches $p = 0.96878$, 0.99686, 0.99960 and 0.99989 at $T = 4, 8, 12, 16$ (and 0.99999 at $24/t$), as fast as Path A and far above the ceiling of Path B, with no shift and no coarse term. Its product-formula version is the fine-chain step with a rescaled interaction angle: 64 second-order steps reproduce the exact-slice overlap at $T = 8/t$ (0.996891 against 0.996861) for $112 + 64 \times 144 = 9\,328$ CNOTs, and 128 steps at $T = 16/t$ give 0.999659 (exact 0.999885) for 18 544 CNOTs. This is the baseline that the refinement paths must beat, the analog of the Hartree–Fock ramp of Ref. [1].

C. Matched-observable comparison

Following Sec. V.C of Ref. [1] we feed the same Trotterized rodeo filter — second order, fixed step δt, $\sigma = 5/\Delta_{\mathrm{eff}} = 8.5/t$, forty Gaussian time sets drawn once and shared by all inputs, up to six cycles — eight inputs: the prolonged coarse ground state, the free-fermion determinant, the pruned and the full UCCSD states, the ramp states at $T = 8/t$ and $16/t$, the Path-B state ($\Lambda = 8t$, $n_s = 256$) and the Path-A state ($n_s = 128$, coarse groups split), and compare the acceptance-weighted ensemble infidelity of the accepted runs against the CNOTs per attempt and per accepted preparation (input plus filter, early termination after the first failed ancilla measurement, the expectation of the cost taken over the time sets). Two implementation details differ from paper I and are forced by the Hubbard norms. The fixed step is diagonalized once, so that n steps are applied exactly as $We^{in\phi}W^\dagger$ and the cycle times are quantized to multiples of δt (on average 68 steps per cycle, 2.4×10^4 CNOTs); the filter then projects on the eigenvector of the step unitary closest to the ground state, whose eigenphase is what the energy scan of the rodeo locates on a device, and we take E_{obj} equal to it rather than to E_0. The Trotter floor of the filter is that eigenvector's infidelity with the exact ground state: 2.6×10^{-3} at $\delta t = 0.25/t$

(the step of paper I, where $E_{\rm obj} - E_0 = 0.09t$ would also have spoiled the acceptance), 5.8×10^{-5} at $\delta t = 0.1/t$ and 3.5×10^{-6} at $0.05/t$; we operate at $\delta t = 0.1/t$, where $E_{\rm obj} - E_0 = 0.013t$. Because the cycle times are quantized, every cycle shares the eigenvectors of the step unitary, so the long-run state of the filter is that eigenvector itself and, with $E_{\rm obj}$ on its eigenphase, the long-run acceptance tends to the input's weight on it instead of decaying; this is the fixed-step analog of the averaged post-selection map of paper I, whose dominant eigenoperator was mixed and whose acceptance decayed at 0.2% per cycle. The sampled infidelities carry bootstrap relative standard errors (over the forty time sets) of up to 18% for the unrefined inputs at $M = 6$ and below 10% for the refined and variational ones; the exact-ensemble values of Eq. (8) for the same inputs lie below the sampled ones once the latter approach the Trotter floor, which Eq. (8) does not contain (for the ramp state at $T = 16/t$: 5.3×10^{-6} exact against 6.2×10^{-5} sampled at $M = 6$).

Tables VI and VII and Fig. 9 give the verdict. At every target the cheapest route is the ordinary adiabatic ramp from the free-fermion determinant at $T = 16/t$: 1.9×10^4 CNOTs for 10^{-3} without any filter cycle, 4.0×10^4 for 3×10^{-4} with one cycle and 9.1×10^4 for 10^{-4} with three. The pruned UCCSD state costs 3.3 and 2.9 times more at the first two targets and does not reach 10^{-4} within six cycles; the full pool is a further 10–25% more expensive for the same accuracy. The two-local refinement path, Path B at $\Lambda = 8t$, costs 7.4 and 4 times the ramp, because its preparation is twice as expensive (3.7×10^4 against 1.9×10^4 CNOTs) and starts twenty times worse ($1 - p = 7.9 \times 10^{-3}$ against 3.4×10^{-4}) owing to the finite-Λ ceiling. Path A delivers the best unfiltered input ($1 - p = 1.8 \times 10^{-4}$) but at 3×10^5 CNOTs, sixteen times the ramp, almost entirely in the conditioned coarse term; it is the only input other than the ramp to reach 10^{-4}, at 3.4×10^5 CNOTs. The prolonged state and the bare determinant, at 16 and 112 CNOTs, need six cycles and 1.5–1.6×10^5 CNOTs to reach 5×10^{-3}, with acceptances of 0.80 and 0.78, and reach none of the targets. Only the inputs that start within a decade of the floor — the ramp at $T = 16/t$ and Path A — converge to it within six cycles; the others are still a factor 1.5 to 80 above it at $M = 6$, so the floor is set by δt but whether it is reached is set by the input.

Three qualifications apply as in paper I. The costs are per accepted preparation at the operating point of each input; the setup costs — the classical optimization and its measurements for UCCSD, the calibration of $(T, \mu_{\rm buf}, n_s)$ or of (T, n_s) for the adiabatic inputs, the energy scan that locates $E_{\rm obj}$ for the filter — are not included and are amortized over the preparations that use them. The refinement parameters are not optimized: $T = 16/t$ and $\mu_{\rm buf} = 2t$ are the first grid values that meet the accuracy criterion, and a joint search over $(T, \mu_{\rm buf}, \Lambda, n_s, \delta t)$ may move every curve by a factor of order unity, but not the factor sixteen between Path A and the ramp, which is the cost of the conditioned coarse term, nor the twenty-fold ceiling of Path B. And the comparison is at one size and one coupling, where the free-fermion determinant has the same overlap and the same $\Delta_{\rm eff}$ as the prolonged coarse state; on larger lattices and at stronger coupling, where the free-fermion overlap with the interacting ground state decays, the ramp loses its starting advantage while the coarse eigenstate, which already contains the correlations at coarse resolution, does not — that is the regime in which the refinement can pay, and it is not reached at sixteen qubits. Refinement is therefore not a gate saver on the Hubbard chain at this size either, and its merit remains that it needs no classical optimizer, no training measurements and no choice of ansatz.

VI. DISCUSSION AND OUTLOOK

a. Lattice versus basis refinement. The lattice refinement of the Hubbard model is structurally cleaner than the basis refinement of the pairing model: a Galerkin coarse model in closed form, a self-similar hierarchy in one dimension, a two-local Hamiltonian with $O(L_f)$ Pauli strings of weight two, a prolongation of $O(N_c)$ gates. Against this stands a coarse term that is conditioned on $2N_c$ qubits, and Path B as its two-local replacement with a $(t/\Lambda)^2$ ceiling and a step count that grows with Λ. The Hubbard path also needs three to ten times more Trotter steps than the pairing path for the same adiabatic time, because the shifted generators have larger norms; this, more than the adiabatic time itself, sets the budget.

b. What the matched comparison shows. The ordinary adiabatic ramp from the free-fermion determinant is the cheapest route to every target accuracy at $U = 4t$ on sixteen qubits, by a factor of three over the pruned variational ansatz, four to seven over the two-local refinement path and sixteen over the projector path. The reason is specific to the size: the determinant has the same overlap with the target (0.78 against 0.80) and the same symmetry-protected gap as the prolonged coarse eigenstate, so the ramp, which pays only for fine-chain steps, starts from the same place and travels at the same speed with cheaper gates. The case for refinement therefore rests on the regime where the coarse eigenstate is a better start than the determinant — larger lattices and stronger coupling, where the free-fermion overlap decays with size while the coarse state carries the correlations at coarse resolution — and that regime must be demonstrated, not assumed; the present comparison fixes the baseline it has to beat. Two further lessons are inherited from paper I. A conditioned coarse term is never affordable: the two-local path, or a schedule that avoids the embedding altogether, is the only form of the projector path worth compiling. And the figure of merit is the cost at fixed accuracy, per accepted preparation, with the filter cycles included; at fixed cycle count every input "converges to the floor" only in the sense that the best ones do.

TABLE VI. Matched-observable comparison for the chain $4 \to 8$ at $U = 4t$: the same Trotterized rodeo filter (second order, $\delta t = 0.1/t$, $\sigma = 8.5/t$, $E_{\rm obj}$ the Trotter-effective ground energy, forty time sets) applied to every input. p is the overlap of the input and "prep" its CNOT count (prolongation 16; Givens network 112; refinement or ramp n_s second-order steps of 144 CNOTs, plus $2 \times 1\,088$ per step for the coarse term of Path A; UCCSD pruned 15 120, full 30 864); $M = 0$ is the unfiltered input. For $M = 2$ and 6 cycles: the ensemble infidelity (bootstrap relative standard error in parentheses), the cumulative acceptance P, and the CNOTs per attempt (att.) and per accepted preparation (acc.).

input	p	prep	$M=0$	$M=2$				$M=6$			
			$1-\mathcal{F}$	$1-\mathcal{F}$	P	att.	acc.	$1-\mathcal{F}$	P	att.	acc.
prolonged	0.7974	16	2.0×10^{-1}	6.0×10^{-2} (7%)	0.848	4.7×10^{4}	5.2×10^{4}	4.7×10^{-3} (18%)	0.802	1.4×10^{5}	1.5×10^{5}
determinant	0.7804	112	2.2×10^{-1}	6.6×10^{-2} (5%)	0.831	4.7×10^{4}	5.3×10^{4}	5.0×10^{-3} (17%)	0.779	1.4×10^{5}	1.6×10^{5}
UCCSD pruned (98)	0.9969	15 120	3.1×10^{-3}	8.2×10^{-4} (6%)	0.997	6.2×10^{4}	6.2×10^{4}	1.1×10^{-4} (7%)	0.997	1.6×10^{5}	1.6×10^{5}
UCCSD full (198)	0.9969	30 864	3.1×10^{-3}	8.1×10^{-4} (6%)	0.997	7.7×10^{4}	7.8×10^{4}	1.1×10^{-4} (7%)	0.997	1.7×10^{5}	1.7×10^{5}
ramp, $T=8$, $n_s=64$	0.9969	9 328	3.1×10^{-3}	7.8×10^{-4} (14%)	0.998	5.6×10^{4}	5.6×10^{4}	8.9×10^{-5} (10%)	0.997	1.5×10^{5}	1.5×10^{5}
ramp, $T=16$, $n_s=128$	0.9997	18 544	3.4×10^{-4}	1.3×10^{-4} (6%)	1.000	6.5×10^{4}	6.5×10^{4}	6.2×10^{-5} (3%)	1.000	1.6×10^{5}	1.6×10^{5}
Path B ($\Lambda=8t$), $n_s=256$	0.9921	36 880	7.9×10^{-3}	2.0×10^{-3} (6%)	0.994	8.3×10^{4}	8.4×10^{4}	2.0×10^{-4} (9%)	0.992	1.8×10^{5}	1.8×10^{5}
Path A (split), $n_s=128$	0.9998	296 976	1.8×10^{-4}	9.1×10^{-5} (2%)	1.000	3.4×10^{5}	3.4×10^{5}	6.0×10^{-5} (1%)	1.000	4.4×10^{5}	4.4×10^{5}

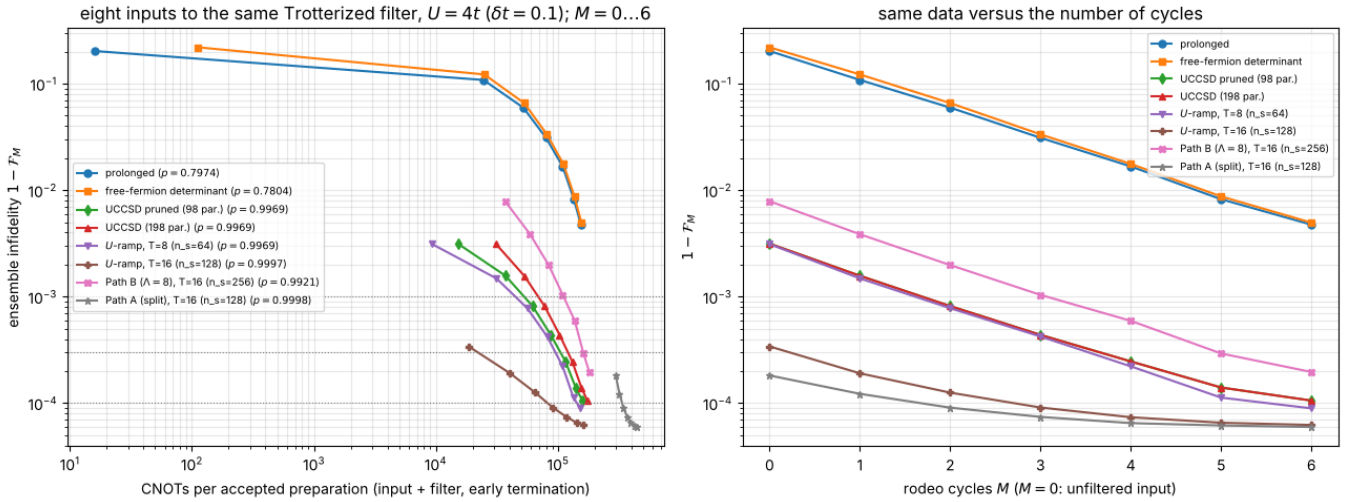


FIG. 9. Cost versus accuracy for the chain $4 \to 8$ at $U = 4t$: ensemble infidelity against CNOTs per accepted preparation (left; the first point of each curve, $M = 0$, is the unfiltered input, dotted lines the targets of Table VII) and against the number of cycles (right) for the eight inputs of Table VI, all cleaned by the same Trotterized filter. The refined inputs reach the Trotter floor of 5.8×10^{-5} within six cycles; the unrefined ones do not.

c. Which gap matters. The quantity that controls the refinement is the symmetry-protected effective gap $\Delta_{\rm eff}$, not the sector gap; this is why T^* is flat in U and in size here, why the sufficient time of the adiabatic theorem is a closer (though still remote) estimate when evaluated with the symmetry-resolved gap, and why $\Delta_{\rm eff}$ is the right width for the rodeo filter. Ref. [1] found that its scan of the refinement time against the shift-controlled gap did not identify a power law (two exponents fit the four points equally well, and the response norm of the path varied with the shift); the lesson here is that any such scan must be made against the gap in the symmetry sector of the tracked state, with the response norm held fixed, before an exponent can be read off. Its converse, the zero-overlap plaquette, is a warning for any refinement or nucleation scheme: the coarse state must carry the quantum numbers of the target, which the ED diagnostics of Sec. II — or, on a device, measurements of $\mathbf{S}^2$ and of the point-group characters — can check before any gate is spent. The singlet–triplet crossing between the half-filled plaquette and the dilute 4×4 lattice is a property of the filling, not of U, over the whole range studied.

d. Coarse accuracy. Composing steps with refined inputs, and starting from a variational coarse state, both show the same ceiling: an admixture of other coarse eigenstates in the input refines to other fine eigenstates and caps the final overlap at roughly the input fidelity. The coarse stage must therefore be as accurate as the target, or the filter must pay for the difference; the filled dimer, which is exact, is from this point of view the best coarse state the hierarchy has.

e. Strong coupling. At fixed U/t the hierarchy is self-similar, so refinement does not by itself reach the

TABLE VII. Operating points: the minimum CNOTs per accepted preparation with which each input of Table VI reaches a target ensemble infidelity, with the number of cycles M that attains it in parentheses ($M \leq 6$; a dash means the target is not reached within six cycles). The refinement inputs are those of Table VI ($T = 16/t$).

input	10^{-3}	$3{\times}10^{-4}$	10^{-4}
prolonged / determinant	—	—	—
UCCSD pruned	$6.2{\times}10^4$ (2)	$1.2{\times}10^5$ (4)	—
UCCSD full	$7.8{\times}10^4$ (2)	$1.3{\times}10^5$ (4)	—
ramp, $T = 8$	$5.6{\times}10^4$ (2)	$1.1{\times}10^5$ (4)	$1.5{\times}10^5$ (6)
ramp, $T = 16$	$1.9{\times}10^4$ (0)	$4.0{\times}10^4$ (1)	$9.1{\times}10^4$ (3)
Path B, $\Lambda = 8t$	$1.4{\times}10^5$ (4)	$1.6{\times}10^5$ (5)	—
Path A	$3.0{\times}10^5$ (0)	$3.0{\times}10^5$ (0)	$3.4{\times}10^5$ (2)

t–J regime; a path that raises U along with the resolution, or a refinement at fixed filling (prolonging with doubled particle number, which is not a state map) are the options, and the ED code's t–J solver gives the reference. The rodeo filter, in contrast, applies unchanged to the t–J Hamiltonian in its constrained sector. [Decide whether a t–J section belongs in this paper.]

f. Relation to nucleation. In one dimension the dimer-seeded hierarchy and the nucleation of a chain from a dimer [14] start from the same state and reach the same target by refining resolution versus growing the domain; the Hubbard chain is the natural place to compare the two paths at matched gate count, and the optimized variants of either (a warm-started variational path) are the obvious way to cut the 10^2–10^3 Trotter steps found here.

g. Hardware targets. The $4 \to 8$ chain (16 qubits) and the $2 \times 2 \to 4 \times 4$ plaquette at $(1, 1)$ (32 qubits, 256-dimensional sector) are within reach of current devices; the Trotter step is shallow (72 CNOTs first order at $L_f = 8$), and Path B needs nothing beyond the fine Hamiltonian's own blocks. [PBC/APBC variants; a schedule with $\Lambda \to \infty$ at $s = 0$; Path-B at $T < 16$ since it saturates early.]

h. Open points. The refinement and filter parameters are calibrated against the exact answer, as in paper I; a device-level protocol (an energy scan for E_{obj}, acceptance statistics for σ and δt, a convergence test in T and n_s on the measured energy) is the missing piece for a hardware run. The variational comparator is a demonstrated restricted ansatz; an adaptive construction, and a comparison at matched shot budget including the training measurements, are not done. The size and coupling dependence of the ramp-versus-refinement comparison — the one measurement that can turn the verdict — requires the $8 \to 16$ chain at the circuit level (14 400 states, Krylov propagation) and is the natural next step.

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CODE AVAILABILITY

All results are produced by the notebook `notebooks/ResolutionRefHubbard.ipynb`, built by `scripts/build_notebook_hubbard.py` on the module `pairinglib.hubbard` (lattices, sector ED, block prolongation, Galerkin coarse model, both paths, exact and Trotterized refinement, the ensemble rodeo formula and its Trotterized version, the symmetry diagnostics, the sector-level UCCSD and the gate counting), which is covered by the test suite of the repository; the figures are extracted from the executed notebook. The ED code (Hubbard and t–J models, chain, square and checkerboard lattices, dense, Lanczos and matrix-free solvers) is the notebook `Hubbard.ipynb` [12], whose conventions the module follows. [Repository link.]

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